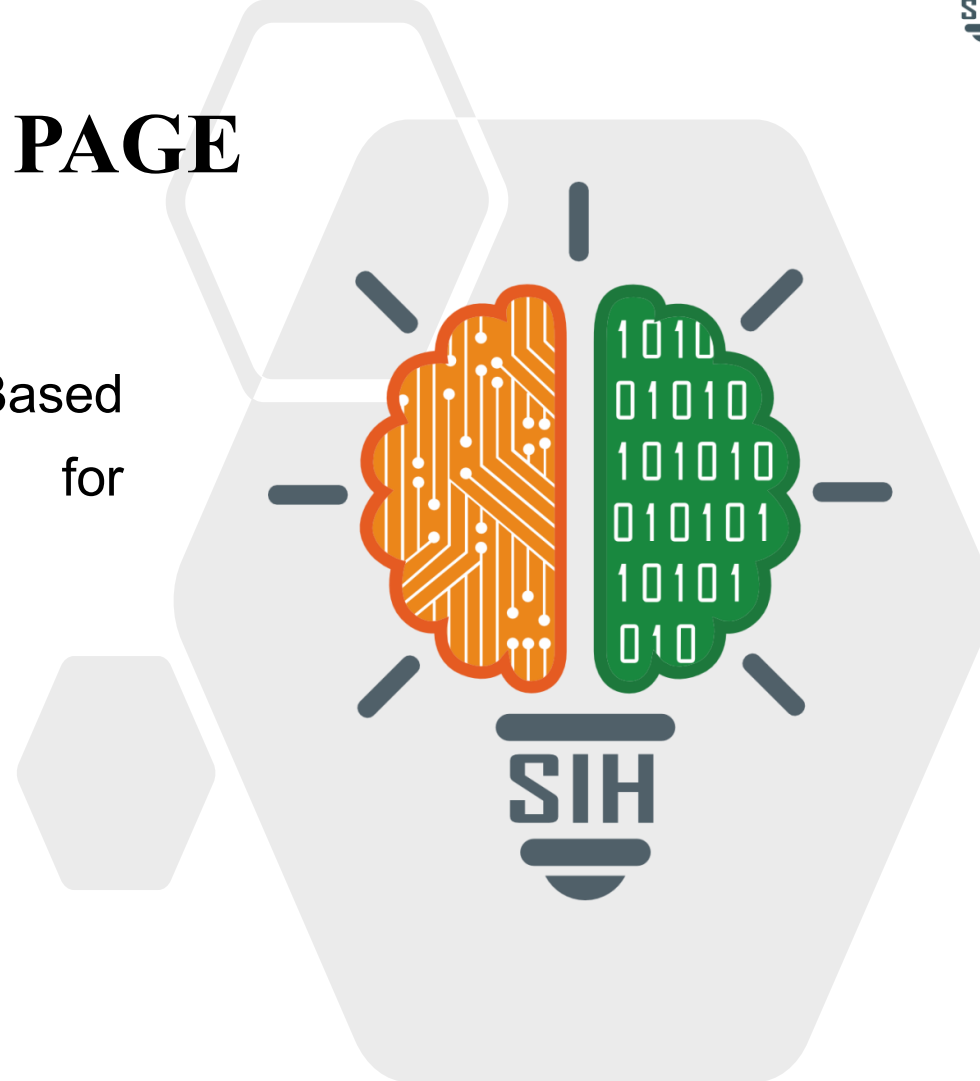


# SMART INDIA HACKATHON 2026



## TITLE PAGE

- **Problem Statement ID – 26168**
- **Problem Statement Title-** AI-ML Based Intelligent Dead Reckoning System for Seamless Navigation
- **Theme-** Smart Vehicles
- **PS Category-** Software
- **Team ID-**137165
- **Team Name – Recursive Minds**

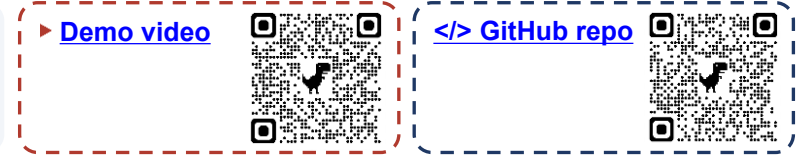


424 % → **12.03 %**  
blackout drift: naive maths → Smart IDR

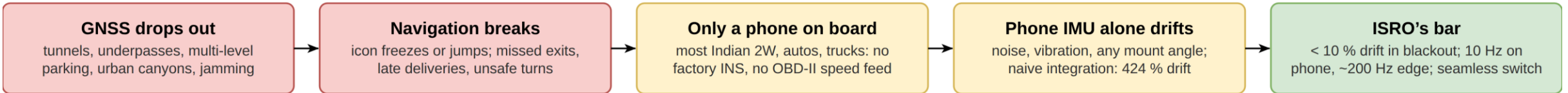
**9.7 %**  
drift on trips unseen in training

**13 ms**  
per 100 ms, budget phone

**₹0**  
extra hardware



## ❖ The problem (PS 26168, ISRO): why navigation fails without GNSS



## ❖ Proposed Solution

A software engine that turns any smartphone in a dashboard cradle or handlebar mount into a self-contained GNSS + INS navigator. **No OBD-II port, no wheel sensor, no network.** AI reads speed from vibration; physics and the road map keep it honest. The full pipeline runs standalone on a budget phone in airplane mode.

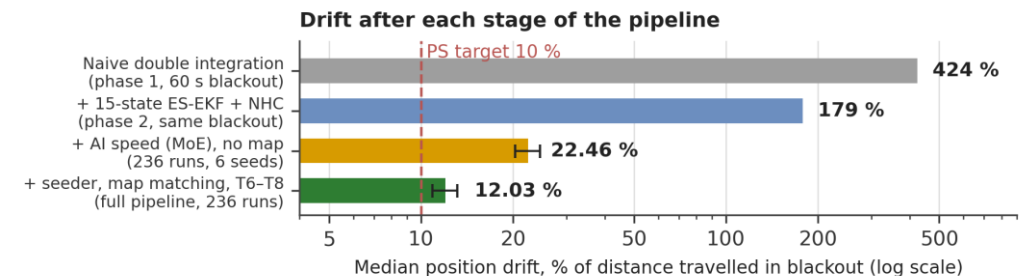
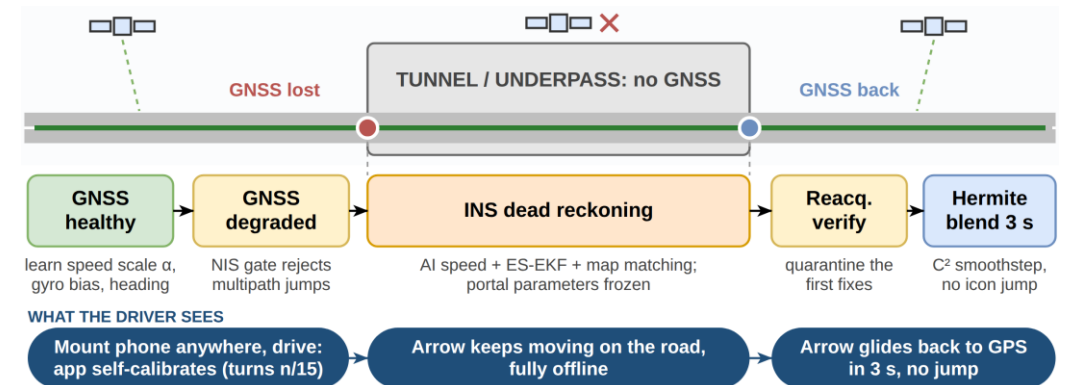
## ❖ How it addresses the problem

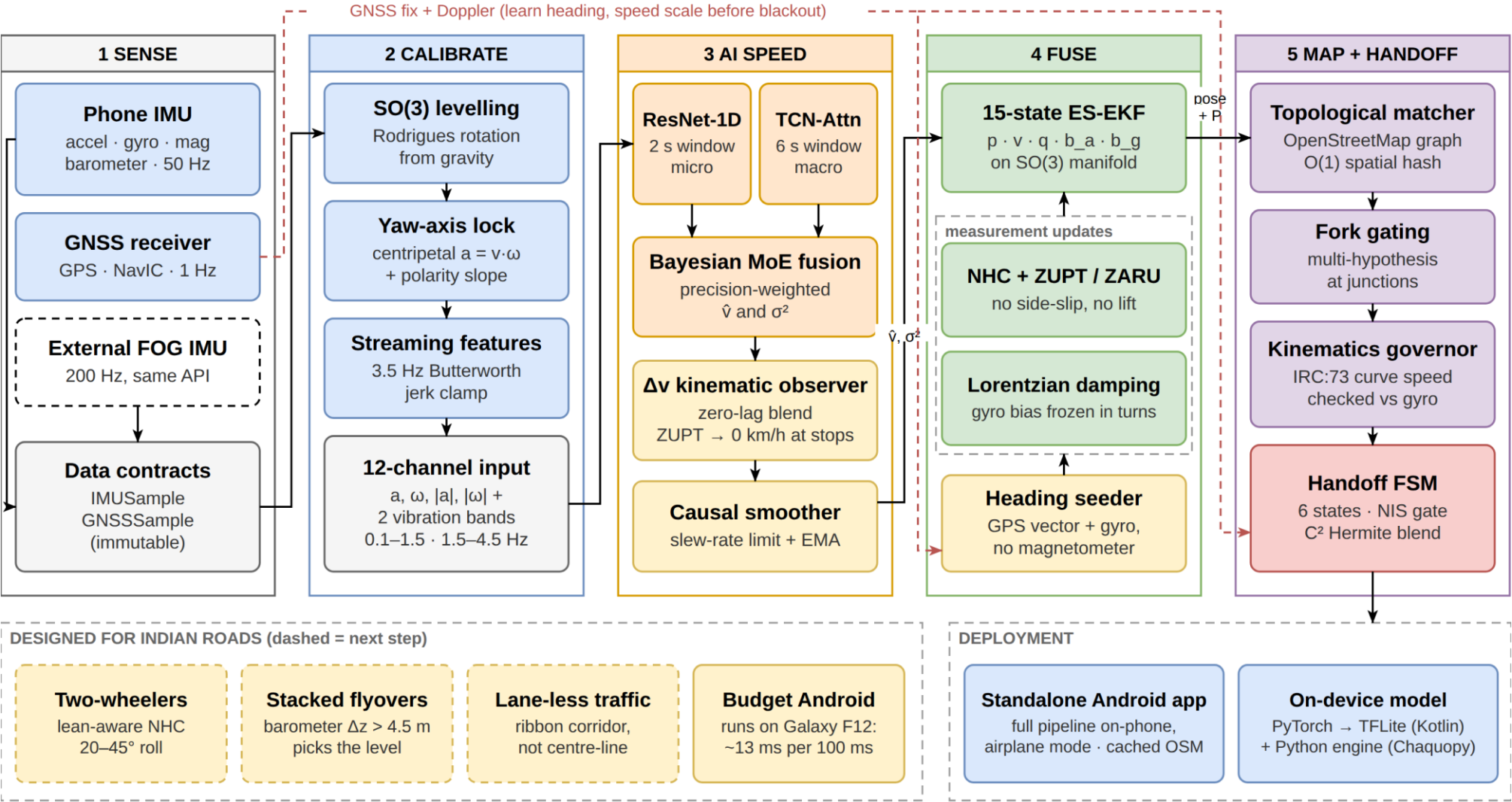
- **No speedometer:** a Bayesian mixture-of-experts reads 12 IMU channels and predicts forward speed, trained on 10 Hz CAN wheel speed.
- **Any phone mount:** SO(3) gravity levelling and centripetal yaw lock align the phone to the vehicle, and keep re-checking.
- **Drift and wrong roads:** 15-state ES-EKF with non-holonomic constraints and ZUPT, then topological map matching on OpenStreetMap.
- **Jumping icon:** a 6-state handoff machine eases GNSS back in over 3 s.

## ❖ Innovation & uniqueness

- **GPS-vector heading seeder** replaces the magnetometer, which is off by 28–76° inside a steel cabin.
- **Kinematic  $\Delta v$  observer** removes the 1.3 s lag window-based speed models have when braking.
- **Designed for Indian roads:** any mount on cars, autos and 2W; lean-aware NHC, barometer flyover gating and lane-less corridors are our next steps.
- **Leak-free evaluation:** tuned on dev seeds, confirmed on unseen held-out seeds.

## ❖ How a GNSS blackout is handled, and what the driver sees





| TECH STACK                                                                                    |                                                                                                       |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <br>Python | <br>PyTorch        |
| <br>NumPy  | <br>SciPy          |
| <br>Kotlin | <br>Android        |
| <br>C++17 | <br>OpenStreetMap |
| <br>ONNX | <br>TFLite       |

❖ **Feasibility: built and tested, not just planned**

- **Engine runs end to end** on a laptop and standalone on the phone (Python via Chaquopy + 2.5 MB TFLite model); phone output = laptop output to 0.000 m.
- **Android app on a Galaxy F12, airplane mode:** ~13 ms per 100 ms step, cached OSM map, CSV logging, benchmark replay, GNSS handoff.
- **5 real IO-VNBD trips** with CAN wheel-speed truth; 160 automated tests, incl. future-leak and phone-parity tests.
- **Viable at scale:** zero hardware cost, one model for every vehicle class; new data needs fine-tuning only.

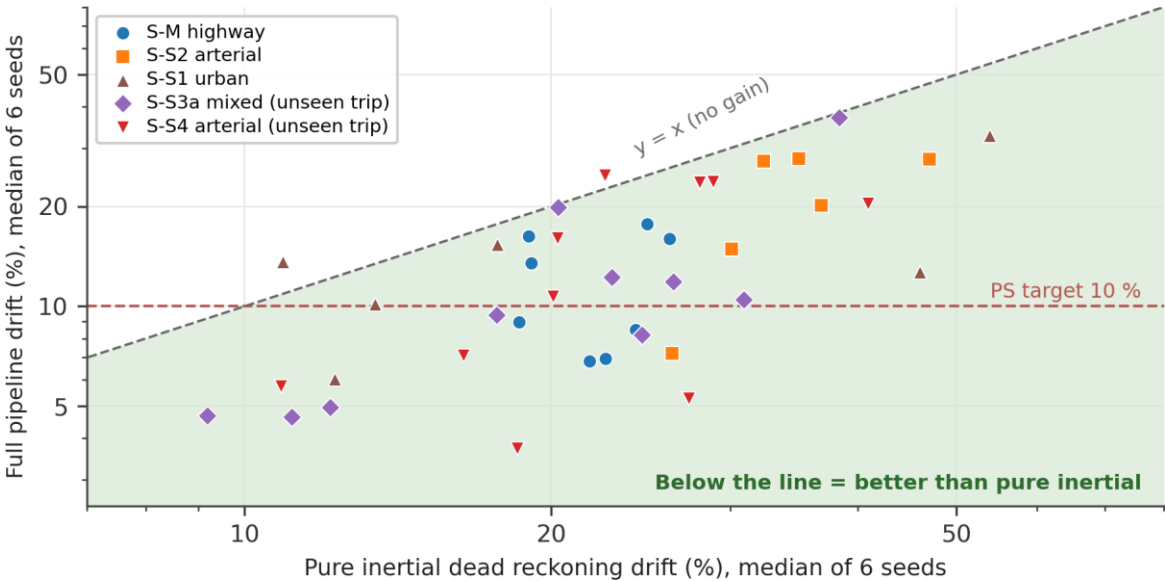
❖ **Challenges we hit while building, and how we fixed them**

| What we assumed                               | What real data showed                                  | What we built                              |
|-----------------------------------------------|--------------------------------------------------------|--------------------------------------------|
| Neural net can learn heading from the IMU     | Cabin steel bends the compass by 28–76°                | Physics heading fusion + GPS-vector seeder |
| Phone GPS speed is good ground truth          | Phone GPS speed lags by up to 9 s                      | 10 Hz CAN wheel-speed supervision          |
| Neural speed can feed the integrator directly | 1.3 s window lag when braking                          | Kinematic Δv observer, zero phase lag      |
| OSM polylines give true road curvature        | Waypoint kinks cause 6.2 m/s <sup>2</sup> false spikes | Curvature governor, gyro-checked (IRC:73)  |

❖ **Risks ahead, and how we already handle them**

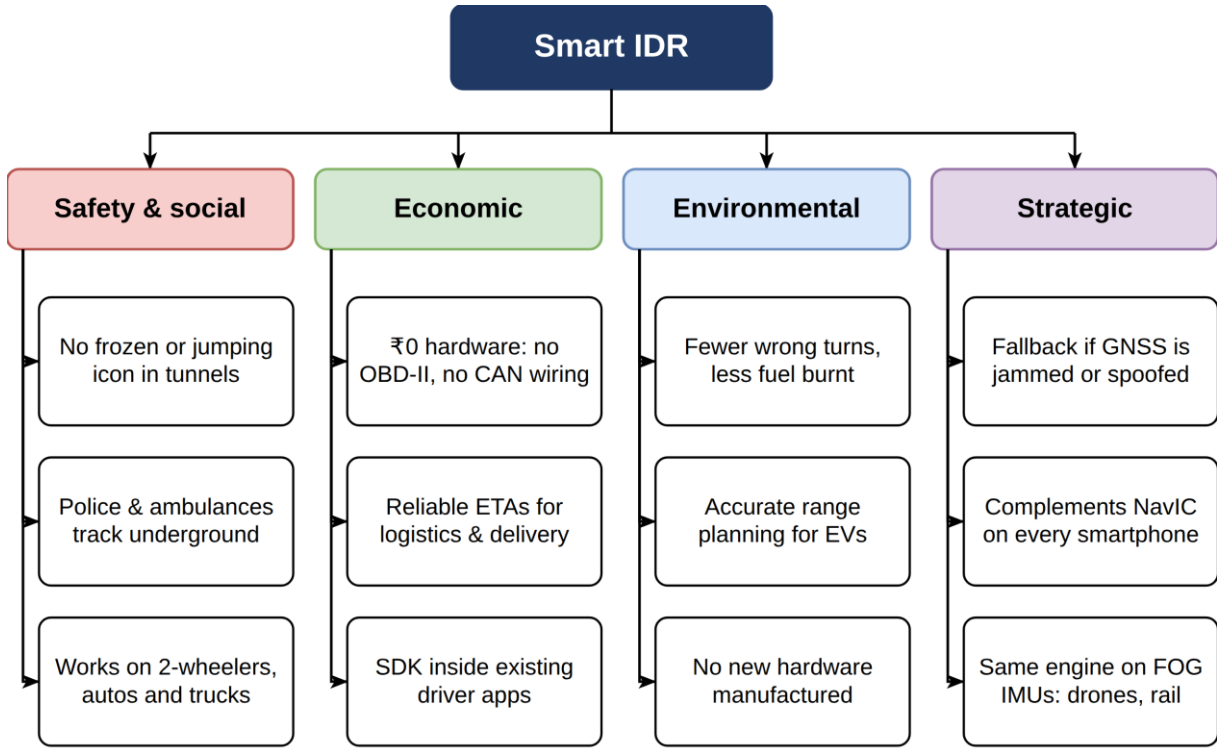
| Potential risk                           | How Smart IDR handles it                                                                              |
|------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Trained on UK trips; Indian roads differ | ✓ <b>Speed scale re-learnt from GNSS before every blackout</b> · next: Indian OBD drives to fine-tune |
| Crawl traffic, stop-and-go               | ✓ <b>ZUPT + Δv observer hold speed at stops</b> · next: crawl-specific tuning                         |
| Phone shifts in its mount mid-trip       | ✓ <b>Mount re-checked on turns; live status on screen</b>                                             |
| No map for a new area                    | ✓ <b>Area prefetched once, cached; works in airplane mode</b> · next: auto-download on route          |
| Battery and heat on budget phones        | ✓ <b>~13 ms per 100 ms step on a Galaxy F12</b> · next: INT8 model                                    |

**Results: 40 blackout scenarios × 6 seeds (236 runs), 5 trips**



38 of 40 scenarios beat pure inertial; 15 finish under 10 % drift, 38 under 30 % (6-seed medians).

| Regime (from the PS)                          | PS target    | Smart IDR             | Status |
|-----------------------------------------------|--------------|-----------------------|--------|
| <b>Tier 1: crawl, &lt; 20 km/h (n = 43)</b>   | < 10 m error | 15.8 m median         | Near   |
| <b>Tier 2: city, 20–50 km/h (n = 152)</b>     | < 15 % drift | <b>12.16 % median</b> | Met    |
| <b>Tier 3: highway, &gt; 50 km/h (n = 41)</b> | < 10 % drift | 13.98 % median        | Near   |
| <b>All scenarios, 236 runs, 6 seeds</b>       | < 10 % drift | 12.03 % median        | Near   |
| <b>Held-out check, 120 runs, 3 seeds</b>      | < 10 % drift | 11.15 % median        | Near   |



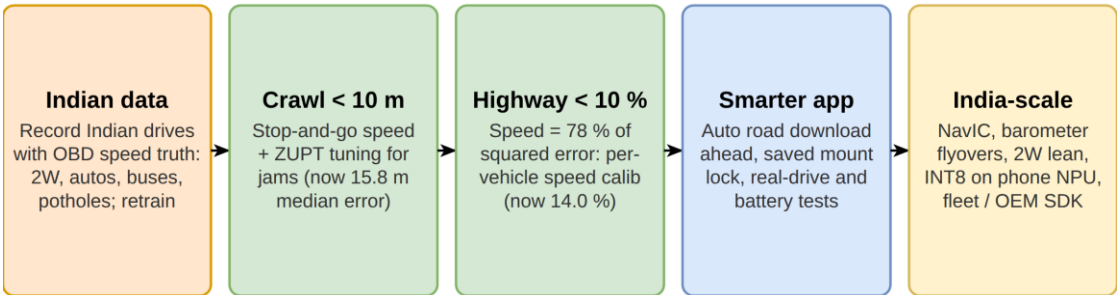
## ❖ How we compare with what exists today

| Capability                     | Google Maps / Mapps | Factory INS    | Smart IDR         |
|--------------------------------|---------------------|----------------|-------------------|
| Tracks through a GNSS blackout | X freezes, jumps    | ✓              | ✓                 |
| Any vehicle: 2W, auto, truck   | ✓                   | X premium cars | ✓                 |
| Extra hardware                 | none                | OEM-fitted     | none              |
| Arbitrary phone mount          | n/a                 | fixed install  | ✓ self-calibrates |
| Published blackout accuracy    | none                | not disclosed  | 12.0 % median     |

## ❖ Potential impact on the target audience

- **2W, autos, trucks, older cars** get navigation without a factory INS.
- **Drift cut from 424 % to 12.03 %** (naive integration → full system); AI + EKF alone 22.46 %; held-out 11.15 %, unseen 9.66 %.
- **Logistics, ride-hailing, emergency fleets** keep tracking in tunnels and basements.
- **Strategic:** survives GNSS jamming, complements NavIC, ports to FOG IMUs.
- **Sustainable delivery:** free driver app; paid SDK for fleets and OEMs funds updates.

## ❖ Future improvements: closing the gaps we measured



❖ **References (links are clickable)**

[1] Brossard et al., AI-IMU Dead-Reckoning, IEEE T-IV 2020 [arxiv.org/abs/1904.06064](#)

[2] Chen et al., IONet, AAAI 2018 [arxiv.org/abs/1802.02209](#)

[3] Herath et al., RoNIN, ICRA 2020 [arxiv.org/abs/1905.12853](#)

[4] Liu et al., TLIO: Tight Learned Inertial Odometry, IEEE RA-L 2020 [arxiv.org/abs/2007.01867](#)

[5] Kendall & Gal, Uncertainties in Bayesian Deep Learning, NeurIPS 2017 [arxiv.org/abs/1703.04977](#)

[6] Newson & Krumm, HMM Map Matching, ACM SIGSPATIAL 2009 [doi.org/10.1145/1653771.1653818](#)

[7] Yang & Gidófalvi, Fast Map Matching, IJGIS 2018 [doi.org/10.1080/13658816.2017.1400548](#)

[8] Onyekpe et al., IO-VNBD dataset, Data in Brief 2021 [doi.org/10.1016/j.dib.2021.106885](#)

[9] Solà, Quaternion kinematics for the error-state KF [arxiv.org/abs/1711.02508](#)

[10] Indian Roads Congress, IRC:73 Geometric design of rural highways

[11] ISRO, NavIC (IRNSS) programme [isro.gov.in/IRNSS\\_Programme.html](#)

[12] SIAM, 2024 domestic sales: 19.5 M of 25.5 M vehicles are 2W [siam.in \(press release\)](#)

[13] OpenStreetMap (roads, ODbL) · ISRO Bhuvan (Indian GIS layers, next) [openstreetmap.org](#)

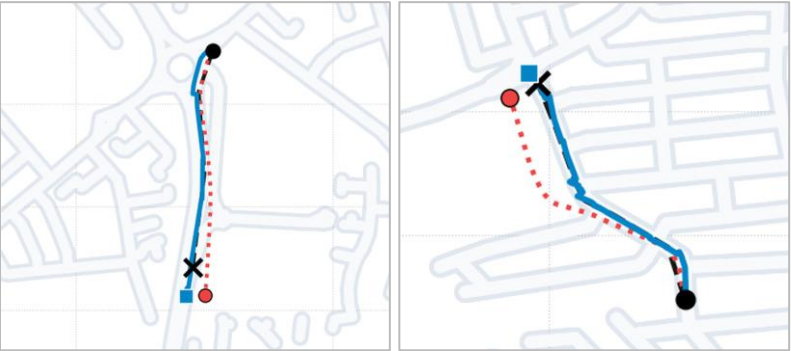
Code studied: mbrossar/ai-imu-dr · Sachini/ronin · cyang-kth/fmm

Our repo: <https://github.com/Recursive-Minds/manas-sih>

❖ **Closest prior work, and what we built on it**

| Prior work                | What it does                                                                                  | Gap for our problem                    | Used in our engine                                    |
|---------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------|
| AI-IMU DR [1]             | Car on KITTI, high-grade IMU bolted in; learns EKF noise                                      | Not a phone; no map, no GNSS handoff   | 15-state ES-EKF + NHC                                 |
| IONet · RoNIN [2,3]       | Phone IMU → walking velocity                                                                  | Pedestrian gait, not vehicle vibration | Bayesian MoE speed                                    |
| TLIO [4]                  | Headset IMU; learned motion + EKF                                                             | Pedestrians only                       | MoE speed + ES-EKF                                    |
| Kendall & Gal [5]         | Uncertainty from deep nets                                                                    | —                                      | MoE speed $\sigma^2$ feeds the EKF                    |
| HMM · FMM [6,7]           | Map-matches GNSS traces                                                                       | Needs GNSS fixes                       | Topological matcher on the DR track                   |
| IO-VNBD [8] · IRC:73 [10] | Car data with CAN speed · road geometry                                                       | UK roads; Indian data next             | MoE training, 356-run benchmark · kinematics governor |
| Smart IDR (ours)          | Phone in any mount, any vehicle, offline: AI speed + ES-EKF + map + 3 s handoff in one engine |                                        |                                                       |

❖ **Real benchmark trajectories (current system, seed 541098)**



#06 Highway · 427 m · 1.9 %

#33 Arterial · 443 m · 4.5 %

Median drift: 12.03 % (236 runs), 11.15 % (120 held-out).

— SMART IDR

our solution

... PURE INERTIAL

baseline, no map

— GROUND TRUTH

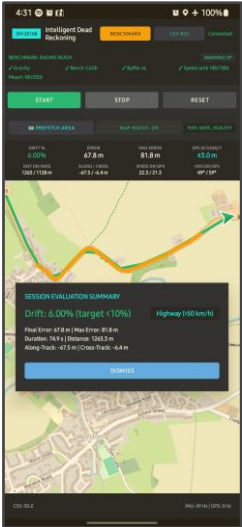
real GPS path

❖ **Our app running standalone on a real phone**



**Live mode on the IIIT Allahabad campus**

Airplane mode, no laptop. Warm-up checks on the phone: gravity, turns n/15, 6 s buffer, speed calibration.



**Benchmark replay #23 on the phone**

1,128 m blackout: 6.0 % drift (67.8 m). Track (orange) vs truth (green); same result as the laptop to 0.000 m.